

TANISHA AGARWAL

tanisha.agarwal@duke.edu | 984-314-6606 | Durham, NC | [LinkedIn/tanisha-agarwal5](https://www.linkedin.com/in/tanisha-agarwal5) | [Portfolio](#)

EDUCATION

DUKE UNIVERSITY, THE FUQUA SCHOOL OF BUSINESS

Durham, NC

MS in Quantitative Management: Business Analytics, Strategy

May 2026

Relevant Coursework: Modern Analytics (Python), Data Infrastructure (SQL), Data Visualization (Tableau), Data Science for Business (R), Decision Analytics & Modeling (Excel), Empirical Economics, Strategic Management

MANIPAL INSTITUTE OF TECHNOLOGY, MAHE

Manipal, India

B.Tech in Electronic and Communications Engineering, Minor in Business Management

July 2024

WORK EXPERIENCE

LAVELLE NETWORKS PVT LTD

Bengaluru, India

Business Analyst

September 2024 - June 2025

- Retained **\$1.4M in revenue** by deploying an automated tracking system for contract renewal that flagged at-risk accounts monthly, enabling proactive outreach to clients due for renewal.
- Reduced reporting errors by automating the reporting ecosystem using **Python, SQL and Excel**, centralizing fragmented data streams to enhance data quality.
- Built PowerBI dashboards on Sales, Profit Margin, and Inventory Turnover that enabled leadership to replace monthly performance reviews with weekly check-ins on live metrics.
- Deployed statistical models to forecast business performance with **90% MAPE on monthly revenue forecasts**, enabling optimized inventory positioning and reduced fulfilment delays.
- Redesigned the client hardware replacement workflow from a reactive, manual process to a proactive, data-triggered operating model by deploying a low-code automation solution on Zoho Creator, using usage-trend analytics to drive replacement decisions and reduce client downtime.

SENSORISE DIGITAL SOLUTIONS PVT LTD

Gurugram, India

Business Analyst Intern

January 2024 - June 2024

- Delivered cross-sector market analysis across 5+ verticals, directly informing the company's portfolio expansion strategy into new markets.
- Led the redesign of the 'KYC SenseTouch' management platform, migrating the backend from **Java to Python/Django** to improve system scalability and long-term maintainability.

ARCADIS-IBI GROUP

Gurugram, India

ICT Intern

July 2023 - August 2023

- Evaluated smart city technical trends for DMIC-IITGNL, optimizing project proposals and **reducing development lead times**.
- Synthesized technical concepts into executive presentations, **improving stakeholder approval speed by 15%**.

GRANT THORNTON BHARAT LLP

Gurugram, India

Public Sector Consultant Summer Trainee

June 2023 - July 2023

- Consolidated data from **20+** disparate sources to deliver market research reports that guided state-level strategic planning and policy decisions.
- Assisted in drafting **cybersecurity frameworks** for state-level initiatives, ensuring alignment with national security standards.

SELECTED PROJECTS

Capstone Project - Genuine Parts Company (NAPA). Led a 5-person engagement with NAPA's pricing organization: built RFM customer segmentation and benchmarked logistic regression, gradient boosting, and XGBoost churn models to identify which pricing strategies; translated findings into a tiered pricing playbook and roadmap delivered to executive stakeholders.

Public Safety Demand Analysis (Tableau, Python, R). Analyzed police, fire, construction and traffic data for the city of Cary, using Tableau, identified geographic crime concentration and seasonal crash risk patterns to guide district-level resource allocation.

Classifying Real vs. AI-Generated Images (Python, Pytorch). Built a custom CNN and fine-tuned ResNet-18 in PyTorch to classify 57K images; achieved 99.9% train accuracy with a GPU-optimized preprocessing pipeline.

Predicting House Prices (Python, Pytorch). Implemented deep MLP models with regularization and dropout to achieve 0.091 RMSE; performed error bias analysis to inform iBuyer pricing strategy and identify systemic valuation risks.

Agentic Workflow for Contract Renewal Automation (LangGraph, Claude API, Python). Designed and built a multi-agent MVP that autonomously processes contracts, extracts renewal terms, scores account churn risk, and drafts stakeholder outreach, demonstrating zero-based redesign of a manual enterprise workflow into an AI-native operating model.

TECHNICAL SKILLS

Languages: Python (PySpark, NumPy, Pandas), R, SQL; **Visualization:** Tableau, PowerBI, ggplot2, matplotlib, Advanced Excel (functions, lookup, pivot tables); **Machine Learning:** Regression (Linear, Logistic, Lasso), SVM, K-means, Decision Trees, CNNs; **Tools/Frameworks:** ETL, PCA, Hypothesis testing, A/B testing, Time-Series Forecasting; **GenAI & Agentic AI:** LangGraph, Claude API, LLMs, Prompt Engineering, Agentic Workflows